

D896

Differential Equations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An engineer in a test lab that runs on 220-volt equipment models a damped part with $y'' + 3t \cdot y' + y^2 = \sin t$. The model was reviewed by two colleagues before the test. How is this equation classified?

- A. Second-order, linear with variable coefficients.
- B. Nonlinear only because of the $\sin t$ forcing term.
- C. Second-order, nonlinear.
- D. First-order, nonlinear, because the highest power of y is 2.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A bacteria culture grows at a rate proportional to its size: $dP/dt = 0.5P$, with $P(0) = 200$ cells per milliliter. The lab keeps its samples at 37 °C. What is $P(2)$?

- A. $200e \approx 544$
- B. 400
- C. $200e^{0.5} \approx 330$
- D. $200 + e \approx 203$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A quantity satisfies $dy/dt = ky$ and triples in 4 hours. What is k , per hour?

- A. $4 / \ln 3 \approx 3.64$
- B. $\ln 3 / 4 \approx 0.275$
- C. $3/4 = 0.75$
- D. $\ln 4 / 3 \approx 0.462$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A mechanical engineer at a firm that ships to three countries models an undamped spring as $y'' + 9y = 0$ with $y(0) = 2$ and $y'(0) = 0$. The mass is painted red for visibility. Which function solves the problem?

- A. $y = 2e^{(-3t)}$
- B. $y = 2\cos 3t + 2\sin 3t$
- C. $y = 2\cos 3t$
- D. $y = 2\cos 9t$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

5.

Wildlife managers at a park with 12 rangers model a deer herd with $dP/dt = 0.3P(1 - P/500)$. Last winter was unusually mild. Which statement about the equilibria is correct?

- A. $P = 0$ is stable and $P = 500$ is unstable.
- B. $P = 500$ is stable and $P = 0$ is unstable.
- C. $P = 0.3$ is the only stable equilibrium.
- D. There is no equilibrium, because a population with positive growth always keeps rising.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

6. [TRANSFER — reversed stem / different variable]

For $dy/dt = f(y)$, the equilibrium $y^* = 2$ is asymptotically stable and $f'(2) \neq 0$. Which must be true?

- A. $f(2) = 0$ and $f'(2) < 0$
- B. $f(2) > 0$
- C. $f''(2) < 0$
- D. $f(2) = 0$ and $f'(2) > 0$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

7.

An analyst who has published four papers on predator-prey models studies $x' = Ax$ with $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Her code runs in under a second. How does the origin behave?

- A. It is a saddle point, so it is unstable.
- B. It is a center, because a symmetric matrix always produces closed orbits.
- C. It is a stable spiral.
- D. It is an unstable node, because every entry of A is positive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

8. [TRANSFER — reversed stem / different variable]

In a phase portrait for $x' = Ax$, every trajectory spirals inward to the origin. Which pair of eigenvalues could A have?

- A. -2 and -3
- B. $2 \pm 3i$
- C. $\pm 3i$
- D. $-2 \pm 3i$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A controls team with 40 licensed software seats solves $y' + 2y = 0$, $y(0) = 5$, using the Laplace transform. The project is due Friday. What is $Y(s)$?

- A. $40 / (s + 2)$
- B. $1 / (s + 2)$
- C. $5 / (s + 2)$
- D. $5 / (s - 2)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A Laplace-domain solution is $Y(s) = 3 / (s - 4)$. What is $y(t)$?

- A. $3e^{-4t}$
- B. $3\cos 4t$
- C. $3 / (t - 4)$
- D. $3e^{4t}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A circuit designer models two decoupled currents with $x' = Ax$, where $A = \text{diag}(2, -1)$ and $x(0) = (1, 3)$. She ran the model overnight on a laptop. Using the matrix exponential, what is $x(t)$?

- A. $(1 + 2t, 3 - 3t)$
- B. $(3e^{2t}, e^{-t})$
- C. $(e^{2t}, 3e^t)$
- D. $(e^{2t}, 3e^{-t})$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A structural engineer on a bridge-vibration study funded by a state grant must solve $y'' - y = e^t$. The study team meets every Tuesday. Which trial form for the particular solution should she use?

- A. $y_p = Ae^t$
- B. $y_p = Ate^t$
- C. $y_p = At^2e^t$
- D. $y_p = Ae^{-t}$

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

An engineer ~~in a test lab that runs on 220-volt equipment~~ models a damped part with $y'' + 3t \cdot y' + y^2 = \sin t$. ~~The model was reviewed by two colleagues before the test.~~ How is this equation classified?

Target word: order and linearity **Decides:** The highest derivative is y'' and the y^2 term makes it nonlinear.

A	Dropped element	Sees the $3t$ coefficient but overlooks the y^2 term.
B	Adjacent category	A forcing term that depends only on t never affects linearity.
C	Correct	y'' sets the order at two; y^2 is a power of the unknown, so the equation is nonlinear.
D	Confusion set	Confuses the power of y with the order of the highest derivative.

2. Answer A Concept 2

A bacteria culture grows at a rate proportional to its size: $dP/dt = 0.5P$, with $P(0) = 200$ cells per milliliter. ~~The lab keeps its samples at 37 °C.~~ What is $P(2)$?

Target word: value of the solution at $t = 2$ **Decides:** $P = 200e^{(0.5t)}$, so $P(2) = 200e^1$.

A	Correct	$P(2) = 200e^{(0.5 \times 2)} = 200e \approx 543.7$.
B	Adjacent category	Adds 0.5×200 each unit of time: linear growth, not exponential.
C	Dropped element	Leaves out the time $t = 2$ in the exponent.
D	Inverted operation	Adds the exponential factor instead of multiplying by it.

3. Answer B Concept 2 · transfer

A quantity satisfies $dy/dt = ky$ and triples in 4 hours. What is k , per hour?

Target word: solve for the rate constant **Decides:** $e^{(4k)} = 3$, so $k = \ln 3 / 4$.

A	Inverted operation	Divides the time by $\ln 3$ instead of $\ln 3$ by the time.
B	Correct	Taking $\ln$ of $e^{(4k)} = 3$ gives $4k = \ln 3$.
C	Dropped element	Omits the natural log that undoes the exponential.
D	Confusion set	Swaps the growth ratio and the time.

4. Answer C Concept 3

A mechanical engineer ~~at a firm that ships to three countries~~ models an undamped spring as $y'' + 9y = 0$ with $y(0) = 2$ and $y'(0) = 0$. ~~The mass is painted red for visibility.~~ Which function solves the problem?

Target word: solution of the IVP **Decides:** $r^2 + 9 = 0$ gives $r = \pm 3i$, so $y = C_1 \cos 3t + C_2 \sin 3t$; the initial values give $C_1 = 2$, $C_2 = 0$.

A	Adjacent category	Real-root form; the roots here are imaginary, so the motion oscillates.
B	Dropped element	Ignores $y'(0) = 0$; this function has $y'(0) = 6$.
C	Correct	Complex roots $\pm 3i$ give $\cos 3t$ and $\sin 3t$; $y(0) = 2$ and $y'(0) = 0$ leave only $2 \cos 3t$.
D	Wrong stem number	Uses the coefficient 9 as the frequency instead of $\sqrt{9} = 3$.

5. Answer B Concept 4

Wildlife managers at a park with 12 rangers model a deer herd with $dP/dt = 0.3P(1 - P/500)$. Last winter was unusually mild. Which statement about the equilibria is correct?

Target word: stability of each equilibrium **Decides:** $f'(0) = 0.3 > 0$ and $f'(500) = -0.3 < 0$.

A	Inverted operation	Reverses the slope test for f' .
B	Correct	Small herds grow away from 0; herds near 500 return to it.
C	Wrong stem number	0.3 is the growth rate, not a zero of f .
D	Absolute language	Growth slows to zero at the carrying capacity of 500.

6. Answer A Concept 4 · transfer

For $dy/dt = f(y)$, the equilibrium $y^* = 2$ is asymptotically stable and $f'(2) \neq 0$. Which must be true?

Target word: reverse: from stability to the derivative condition **Decides:** Stable requires $f'(y^*) < 0$.

A	Correct	An equilibrium is a zero of f , and a negative slope pushes nearby solutions back.
B	Confusion set	At an equilibrium f equals zero; the sign of f itself says which way solutions move.
C	Adjacent category	Concavity is not the first-derivative stability test.
D	Inverted operation	A positive slope makes the equilibrium unstable.

7. Answer A Concept 5

An analyst who has published four papers on predator-prey models studies $x' = Ax$ with $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Her code runs in under a second. How does the origin behave?

Target word: type and stability of the origin **Decides:** Eigenvalues are 3 and -1 : real with opposite signs.

A	Correct	$\det(A - \lambda I) = (1 - \lambda)^2 - 4 = 0$ gives $\lambda = 3, -1$; opposite signs make a saddle.
B	Absolute language	Symmetric matrices have real eigenvalues, which never give a center.
C	Confusion set	Spirals need complex eigenvalues; these are real.
D	Adjacent category	Stability comes from the eigenvalues, not the signs of the entries.

8. Answer D Concept 5 · transfer

In a phase portrait for $x' = Ax$, every trajectory spirals inward to the origin. Which pair of eigenvalues could A have?

Target word: reverse: from portrait to eigenvalues **Decides:** Inward spiral = complex eigenvalues with negative real part.

A	Confusion set	Real negative eigenvalues give a stable node, with no spiraling.
B	Inverted operation	A positive real part spirals outward.
C	Adjacent category	Purely imaginary eigenvalues give a center with closed orbits.
D	Correct	The imaginary part makes the spiral; the negative real part pulls it inward.

9. Answer C Concept 6

A controls team with 40 licensed software seats solves $y' + 2y = 0$, $y(0) = 5$, using the Laplace transform. The project is due Friday. What is $Y(s)$?

Target word: transformed solution **Decides:** $sY - 5 + 2Y = 0$.

A	Camouflage	Uses the seat count, which has nothing to do with the equation.
B	Dropped element	Drops the initial value $y(0) = 5$ from $\mathcal{L}\{y'\}$.
C	Correct	$(s + 2)Y = 5$ gives $Y = 5/(s + 2)$, which inverts to $5e^{-2t}$.
D	Inverted operation	Flips the sign; this would invert to growth, $5e^{(2t)}$.

10. Answer D Concept 6 · transfer

A Laplace-domain solution is $Y(s) = 3 / (s - 4)$. What is $y(t)$?

Target word: inverse transform **Decides:** $\mathcal{L}\{e^{(at)}\} = 1/(s - a)$ with $a = 4$.

A	Inverted operation	Reads the sign backward; e^{-4t} transforms to $1/(s + 4)$.
B	Adjacent category	$\cos 4t$ transforms to $s/(s^2 + 16)$.
C	Confusion set	Swaps s for t without inverting the transform.
D	Correct	$1/(s - 4)$ is the transform of $e^{(4t)}$; the constant 3 carries through.

11. Answer D Concept 7

A circuit designer models two decoupled currents with $x' = Ax$, where $A = \text{diag}(2, -1)$ and $x(0) = (1, 3)$. She ran the model overnight on a laptop. Using the matrix exponential, what is $x(t)$?

Target word: $x(t) = e^{(At)}x(0)$ **Decides:** $e^{(At)} = \text{diag}(e^{(2t)}, e^{(-t)})$.

A	Adjacent category	Uses $I + At$, only the first-order approximation of $e^{(At)}$.
B	Wrong stem number	Pairs each initial value with the wrong eigenvalue.
C	Inverted operation	Loses the negative sign on the second eigenvalue.
D	Correct	Each component is its initial value times $e^{(\lambda t)}$ for its own eigenvalue.

12. Answer B Concept 8

A structural engineer on a bridge-vibration study funded by a state grant must solve $y'' - y = e^t$. The study team meets every Tuesday. Which trial form for the particular solution should she use?

Target word: correct trial form **Decides:** Roots ± 1 mean e^t already solves the homogeneous equation (a simple root).

A	Dropped element	Ignores the overlap; substituting gives $0 = e^t$.
B	Correct	The forcing matches a simple root, so the trial form is multiplied by t .
C	Confusion set	t^2 is for a repeated root; $r = 1$ is a simple root here.
D	Inverted operation	Matches the other homogeneous root, not the forcing term, and still solves the homogeneous equation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Classifying an ODE: order and linearity	Order is the highest derivative present; an ODE is linear when y and its derivatives appear only to the first power, never multiplied together, with coefficients that depend on t alone.	Calling an equation nonlinear because its coefficients vary with t , or reading a power of y as the order.	“Classify” next to terms like y^2 , $y \cdot y'$, or $\sin y$.
2	Exponential growth and decay as an IVP	$dy/dt = ky$ with $y(0) = y_0$ has the solution $y = y_0 e^{kt}$; k is found from $\ln$ of the growth ratio divided by the time.	Treating proportional growth as a fixed amount added each unit of time.	“Rate proportional to the amount present.”
3	Characteristic equation for constant coefficients	For $ay'' + by' + cy = 0$ solve $ar^2 + br + c = 0$: distinct real roots give e^{rt} terms, a repeated root adds a $t \cdot e^{rt}$ term, and complex roots $\alpha \pm \beta i$ give $e^{\alpha t}(\cos \beta t, \sin \beta t)$.	Using the coefficient itself (9) as the frequency instead of its square root (3), or writing real exponentials for complex roots.	A second-order equation with constant coefficients and initial values.
4	Stability of equilibria in autonomous equations	For $dy/dt = f(y)$, equilibria are the zeros of f ; an equilibrium y^* is stable when $f'(y^*) < 0$ and unstable when $f'(y^*) > 0$.	Assuming the zero equilibrium is always the stable one, or reading stability from the sign of f instead of its slope.	“Which equilibrium is stable?” for a logistic or other autonomous model.
5	Eigenvalues classify the phase portrait	For $x' = Ax$: real eigenvalues of opposite sign give a saddle; both negative a stable node; complex with negative real part a stable spiral; purely imaginary a center.	Judging stability from the signs of the matrix entries instead of the eigenvalues.	A 2×2 matrix and a question about the origin’s behavior.
6	Laplace transforms turn IVPs into algebra	$\mathcal{L}\{y'\} = sY - y(0)$ and $\mathcal{L}\{e^{at}\} = 1/(s - a)$; solve for $Y(s)$, then invert.	Dropping the initial value or flipping the sign in $1/(s - a)$.	“Take the Laplace transform” or a $Y(s)$ to invert.
7	Matrix exponential and fundamental solutions	$x' = Ax$ has solution $x(t) = e^{At}x(0)$; for a diagonal A , e^{At} is diagonal with entries $e^{\lambda t}$, so each component grows or decays at its own eigenvalue.	Replacing e^{At} with the linear approximation $I + At$, or pairing initial values with the wrong eigenvalue.	“Fundamental matrix” or “matrix exponential” with a diagonal or diagonalizable A .
8	Undetermined coefficients and resonance	General solution = homogeneous solution + one particular solution; if the forcing term already solves the homogeneous equation, multiply the trial form by t (by t^2 for a double root).	Guessing Ae^{at} for a forcing e^{at} that already solves the homogeneous equation.	The forcing function matches a root of the characteristic equation.

Course scope (public description): Differential Equations examines the methods and theory of ordinary differential equations (ODEs), focusing on the classification and solution of initial- and boundary-value problems. The course advances to the analysis and solution of linear and nonlinear differential equations, including systems of equations and higher-order differential equations with both constant and variable coefficients. Emphasis is

placed on real-world applications and stability analysis. Students will explore advanced techniques such as matrix exponential, fundamental solution matrices, phase-space analysis, Laplace transforms, and an introduction to partial differential equations (PDEs). This course builds on matrix and vector concepts and prepares students for more advanced mathematics courses.